

ML Assignment 01

Dataset: Titanic Dataset (titanic_data_updated.csv)

Total Marks: 100

Exam Instructions :

1. প্রথমে একটি Google Colab ফাইল খুলবে , এরপর প্রথম cell এ নিজের নাম এবং কোর্সে registration করা ইমেইল দিবে
2. Question wise numbering করে Text cell রাখবে এবং এর নিচে Code cell থাকবে, চেষ্টা করবে একটি code cell এ একটি question উত্তর দেয়ার
3. Google colab এর মধ্যে কোডগুলো করবে
4. এবং সেই ফাইলটি 'Anyone with the link' & 'View' Access দিয়ে ফাইলটির Shareable Link টি সাবমিট করবে।

Question Dataset Link:

<https://drive.google.com/file/d/1aHSSJkDphOCbdGRnnXUDPHvPr08Uwtsk/view?usp=sharing>

Question 1 (10 Marks)

Load the Titanic dataset and display:

- Dataset shape
 - First 10 rows
 - 5 random samples
-

Question 2 (10 Marks)

Perform feature screening on the Titanic dataset by:

- Finding total missing values
 - Finding duplicate rows
 - Removing duplicate rows permanently
-

Question 3 (10 Marks)

Perform statistical profiling of the Titanic dataset and display:

- Dataset information
- Statistical summary of numerical columns

Also write 2 observations from the output.

Question 4 (10 Marks)

Perform univariate categorical analysis on the Survived column by:

- Creating a countplot
- Calculating percentage distribution
- Creating a pie chart

Write 2 insights from the analysis.

Question 5 (10 Marks)

Perform univariate numerical analysis on the Age column by:

- Creating a histogram
- Creating a KDE plot

Write 2 observations from the plots.

Question 6 (10 Marks)

Perform multivariate analysis between Sex and Survived using a countplot with hue.

Also calculate normalized survival ratios using `groupby()`.

Write 2 insights from the analysis.

Question 7 (10 Marks)

Create a barplot showing the relationship between Pclass and Fare.

Write 2 observations from the visualization.

Question 8 (10 Marks)

Create a KDE plot to compare the Age distribution of survived and non-survived passengers.

Write 2 insights from the graph.

Question 9 (10 Marks)

Perform feature engineering and train-test split by:

- Dropping PassengerId, Name, and Ticket columns
- Separating features (X) and target (y)
- Splitting the dataset using `test_size=0.2` and `random_state=42`

Display the shapes of train and test datasets.

Question 10 (10 Marks)

Handle missing values in the Titanic dataset by:

- Imputing missing Age values using `SimpleImputer` with mean strategy
- Imputing missing Embarked values using most frequent strategy
- Imputing missing Cabin values using constant strategy with "Missing"

Finally, check whether any missing values remain.